

File Handling



File Handling

- Here we are going to look at the internal storage.
- Internal storage is the storage of the private data on the device memory.
- By default these files are private and are accessed by only your application and get deleted , when user delete your application.

Write the data to the file

- In order to use internal storage to write some data in the file, call the `openFileOutput()` method with the name of the file and the mode.
- The mode could be private , public e.t.c. Its syntax is given below –

```
FileOutputStream fout = openFileOutput("file name here",  
                                         MODE_WORLD_READABLE);
```

- The method `openFileOutput()` returns an instance of `FileOutputStream`.
- So you receive it in the object of `FileInputStream`.
- After that you can call `write` method to write data on the file. Its syntax is given below –

```
String str = "data";  
fout.write(str.getBytes());  
fout.close();
```

Read the data from the file

- In order to read from the file you just created , call the `openFileInput()` method with the name of the file. It returns an instance of `FileInputStream`. Its syntax is given below –

```
FileInputStream fin = openFileInput(file);
```

- After that, you can call `read` method to read one character at a time from the file and then you can print it.
- Its syntax is given below –

```
int c;  
String temp="";  
while( (c = fin.read()) != -1){  
    temp = temp + Character.toString((char)c);  
}  
  
//string temp contains all the data of the file.  
fin.close();
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
```

```
    android:layout_centerInParent="true"
```

```
    tools:context=".MainActivity">
```

```
    <TextView
```

```
        android:text="Enter the Data"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:layout_centerInParent="true"
```

```
        android:id="@+id/textview" />
```

```
    <EditText
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:layout_centerInParent="true"
```

```
        android:id="@+id/editText" />
```

<Button

```
android:id="@+id/button"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:layout_centerInParent="true"  
android:text="Save" />
```

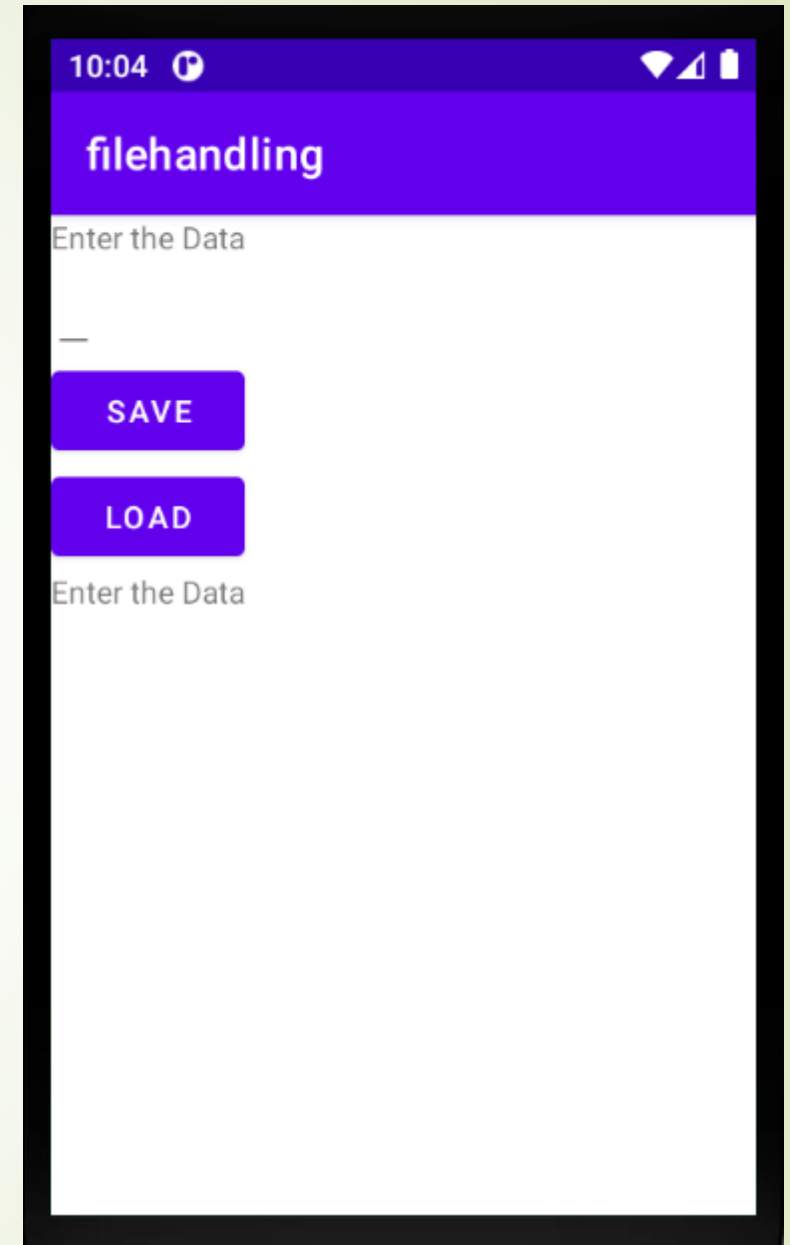
<Button

```
android:id="@+id/button2"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:layout_centerInParent="true"  
android:text="load" />
```

<TextView

```
android:text="Enter the Data"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:layout_centerInParent="true"  
android:id="@+id/textview2" />
```

</LinearLayout>

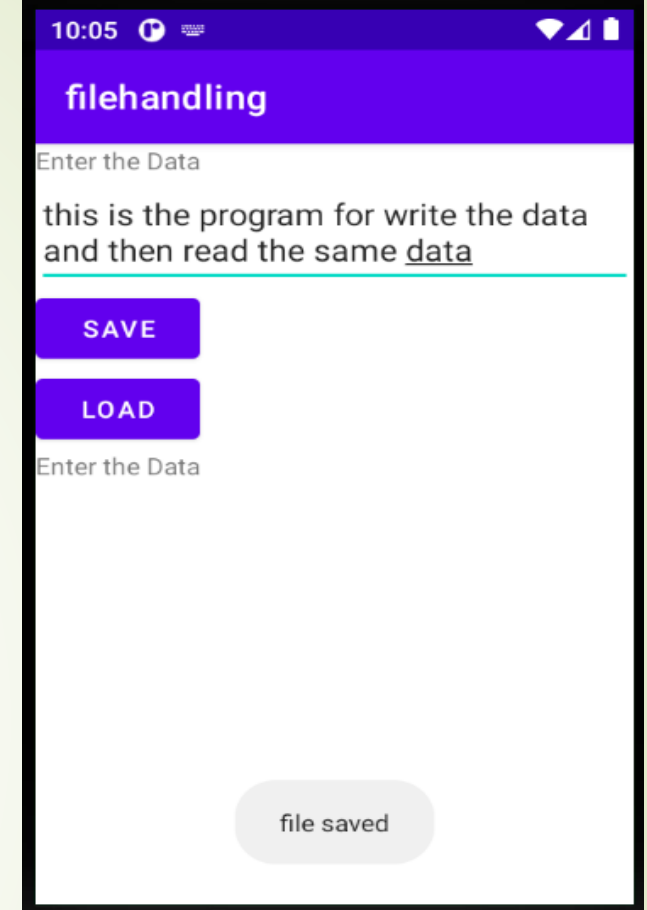




```
package com.example.filehandling;
import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
public class MainActivity extends AppCompatActivity {
    Button b1,b2;
    TextView tv;
    EditText ed1;
    String data;
    private String file = "mydata.txt";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```
b1=(Button)findViewById(R.id.button);  
b2=(Button)findViewById(R.id.button2);  
ed1=(EditText)findViewById(R.id.editText);  
tv=(TextView)findViewById(R.id.textview2);
```

```
b1.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        data=ed1.getText().toString();  
        try {  
            FileOutputStream fOut = openFileOutput(file,MODE_APPEND);  
            fOut.write(data.getBytes());  
            fOut.close();  
            Toast.makeText(getApplicationContext(),"file saved",Toast.LENGTH_SHORT).show();  
        }  
        catch (Exception e) {  
            // TODO Auto-generated catch block  
            e.printStackTrace();  
        } } });
```




```
b2.setOnClickListener(new View.OnClickListener() {
```

```
@Override
```

```
public void onClick(View v) {
```

```
try {
```

```
    FileInputStream fin = openFileInput(file);
```

```
    int c;
```

```
    String temp="";
```

```
    while( (c = fin.read()) != -1){
```

```
        temp = temp + Character.toString((char)c);
```

```
    }
```

```
    tv.setText(temp);
```

```
    Toast.makeText(getApplicationContext(),"file read",Toast.LENGTH_SHORT).show();
```

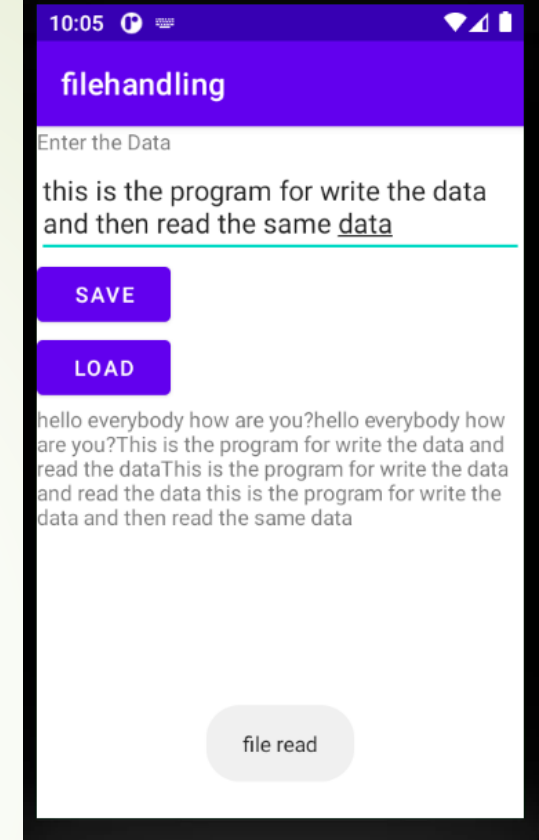
```
}
```

```
catch(Exception e){
```

```
}
```

```
} });
```

```
}}
```



To open the mydata.txt file

➔ device file explorer

➔ data ➔ data

➔ com.example.filehandling

➔ files ➔ mydata.txt

